

Q. 1 (Yogurt). The dining hall serves (and you eat) one flavor of yogurt each day: blueberry (probability 0.2), banana (probability 0.5), and jalapeño (probability 0.3). You have a mild banana allergy: every time you eat banana, you break out in a rash with probability 0.3. The other yogurts do not give you any adverse side effects.

- What is the probability you get a rash on a given day?
- Given that you do *not* get a rash on a given day, what is the probability the dining hall served banana yogurt that day?
- What is the expected number of times during one week (7 days) that you eat banana yogurt, but do not get a rash?

Q. 2 (Snowfall). Let S_P be the snowfall in Princeton and S_N be the snowfall in New York this year (in feet). We model these as nonnegative random variables such that

$$\mathbf{P}\{S_P \geq t, S_N \geq s\} = \frac{1}{t + s + 1}, \quad t, s \geq 0.$$

- What is the probability that more than 1 foot of snow falls in Princeton this year?
- Consider the event that more than 1 foot of snow falls in Princeton, and the event that more than 1 foot of snow falls in New York. Are these two events independent?
- What is the probability that both Princeton and New York get a snowfall of between 1 and 2 feet this year?

Q. 3 (COVID vaccine). It is hard to get an appointment to get the COVID vaccine. For this reason, you are showing up at the vaccination center every day in the hope a slot has opened up at the last minute. The probability that you can get the vaccine in a given day is 0.05, and the availability of a slot in different days is independent. In addition, each day the probability you get exposed to COVID is 0.01, independently for each day and of the availability of vaccination slots.

- What is the probability that in the first week (7 days), you do not get the vaccine and get exposed to COVID three times?
- You need to get two doses of the vaccine for the vaccine to be fully effective. What is the expected day on which you get your **second** dose?
- What is the probability you get exposed to COVID at least once before you have received the **first** dose of the vaccine? (If you get exposed to COVID on the same day that you get the vaccine, you can assume that you got the vaccine first.)